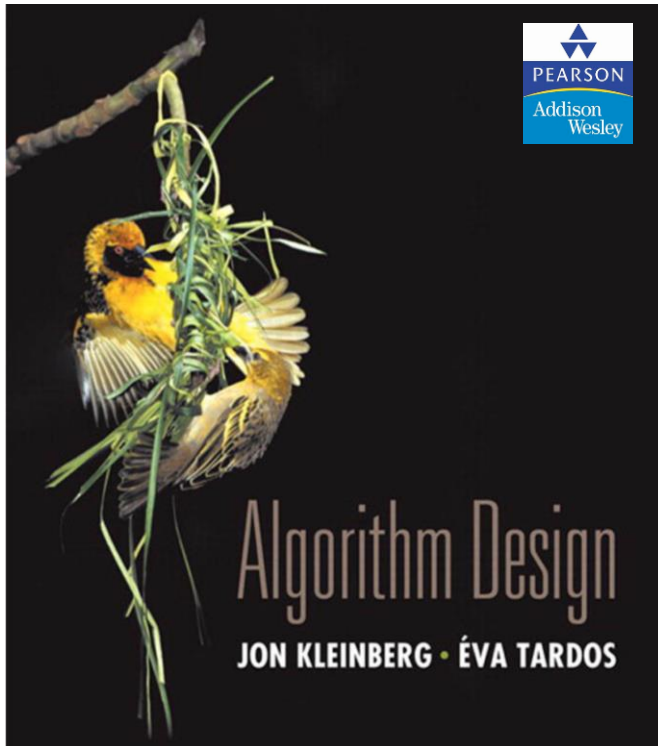




SECTION 11.2

11. APPROXIMATION ALGORITHMS

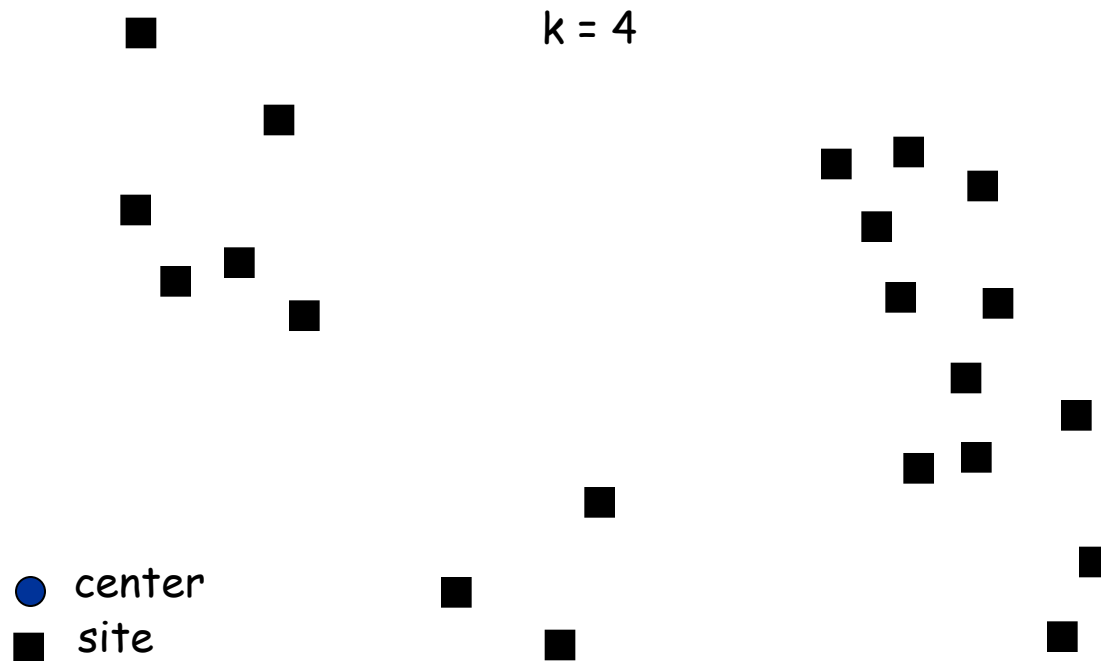
- *load balancing*
- *center selection*

k-Center Problem

k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

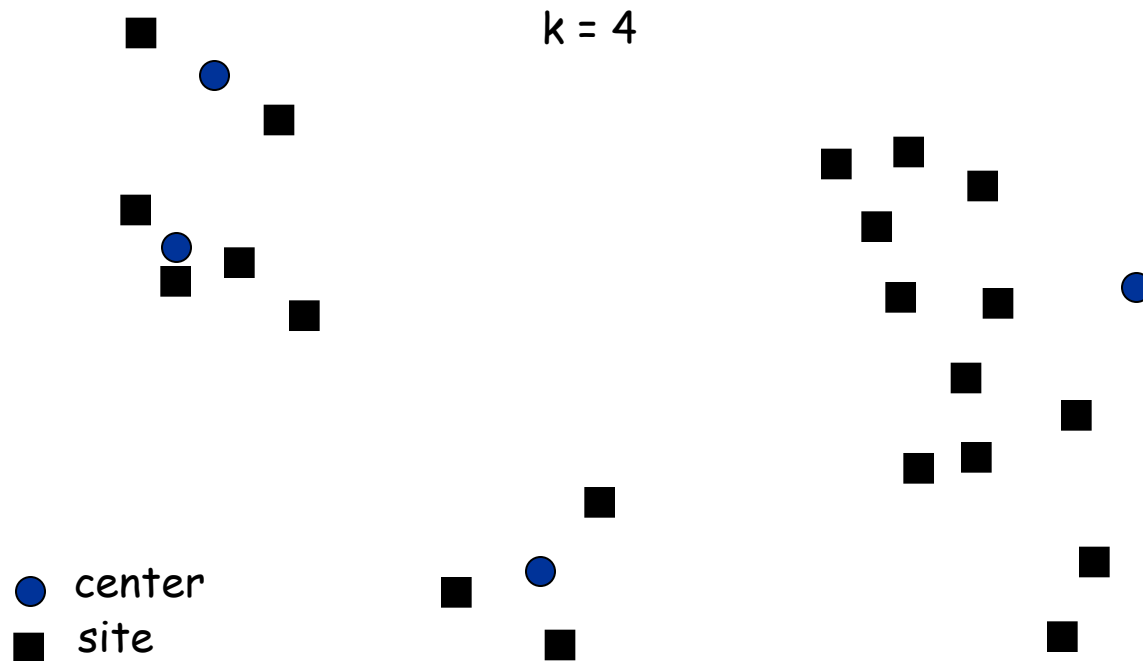
Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

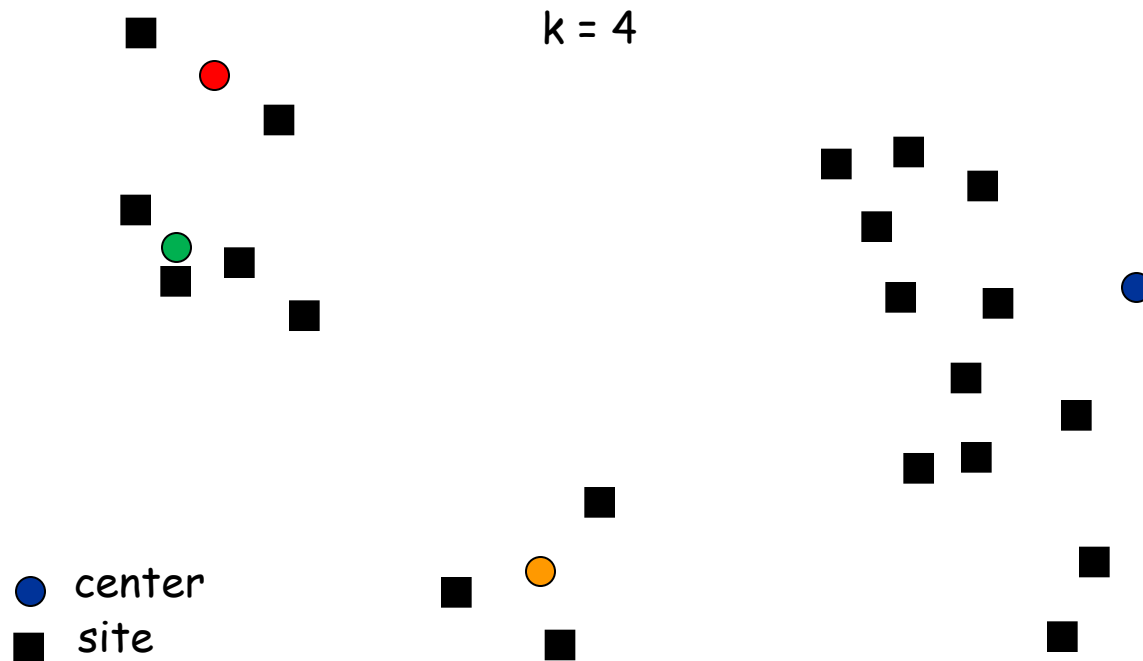
Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

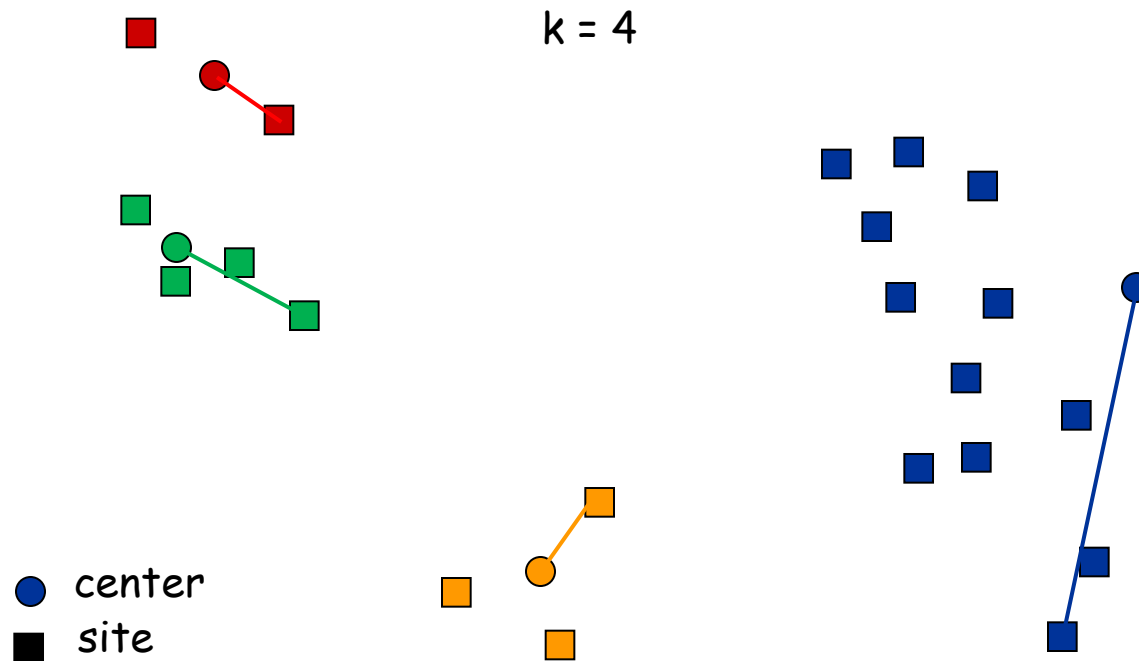
Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

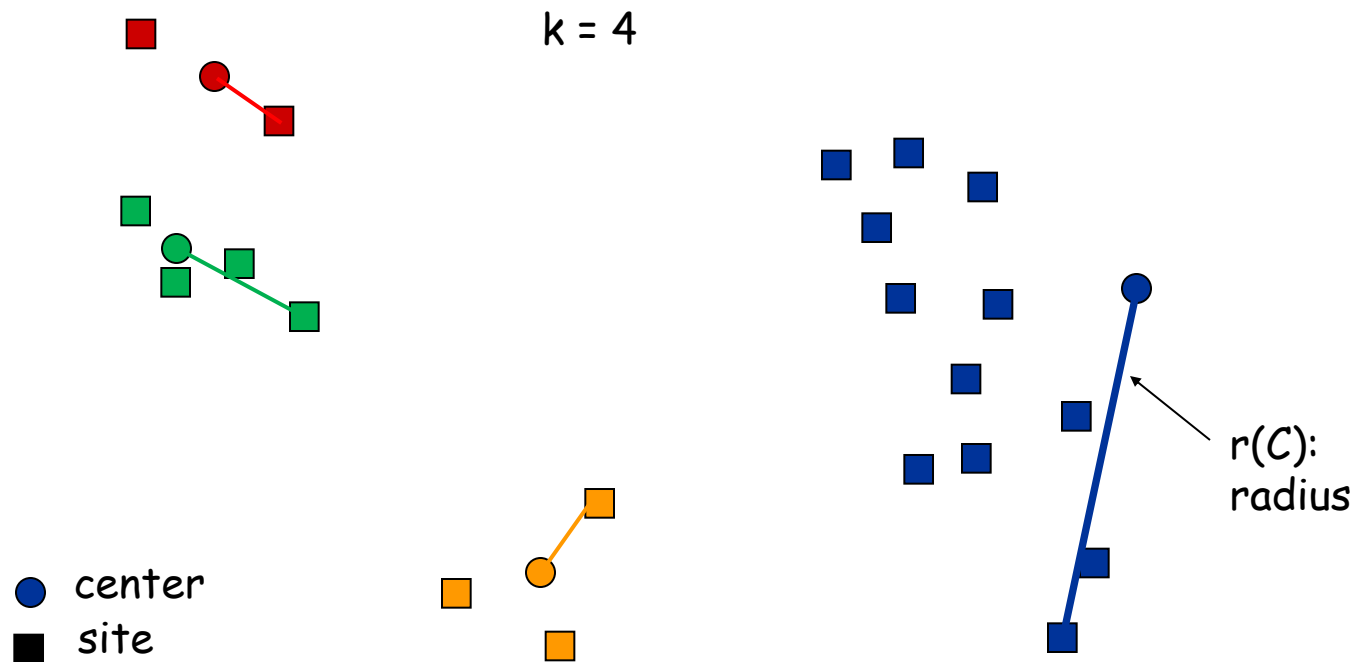
Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

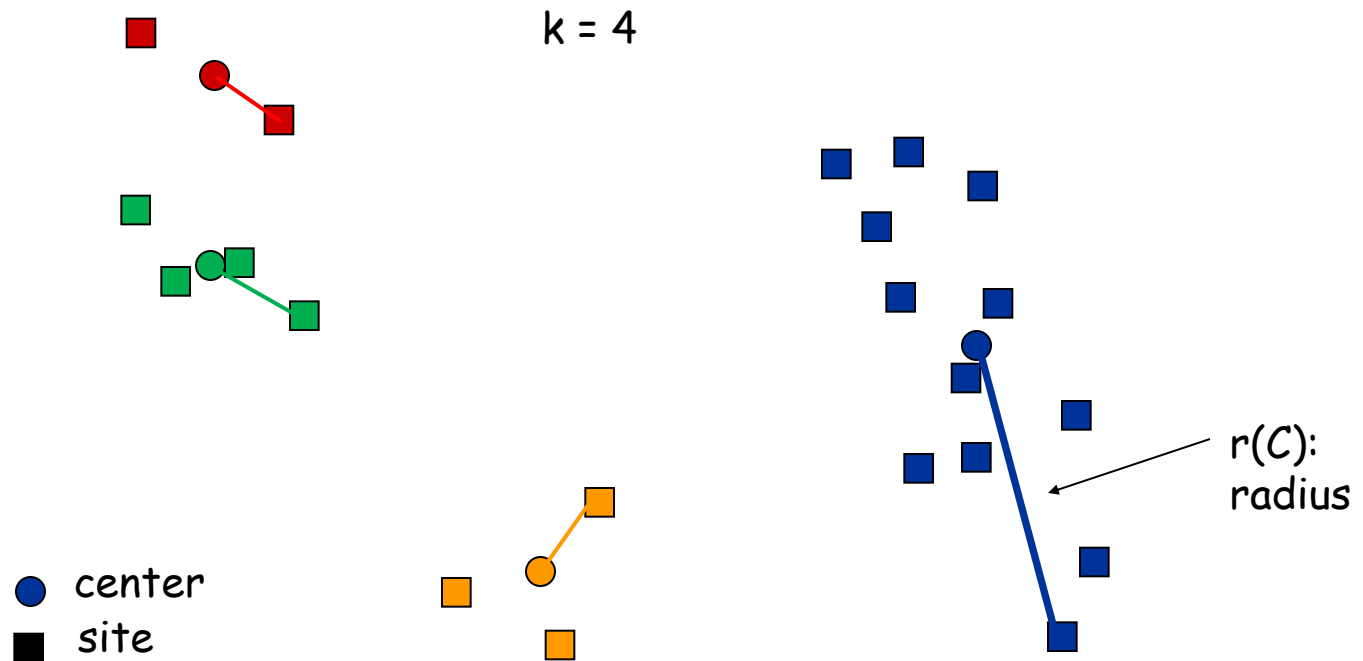
Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

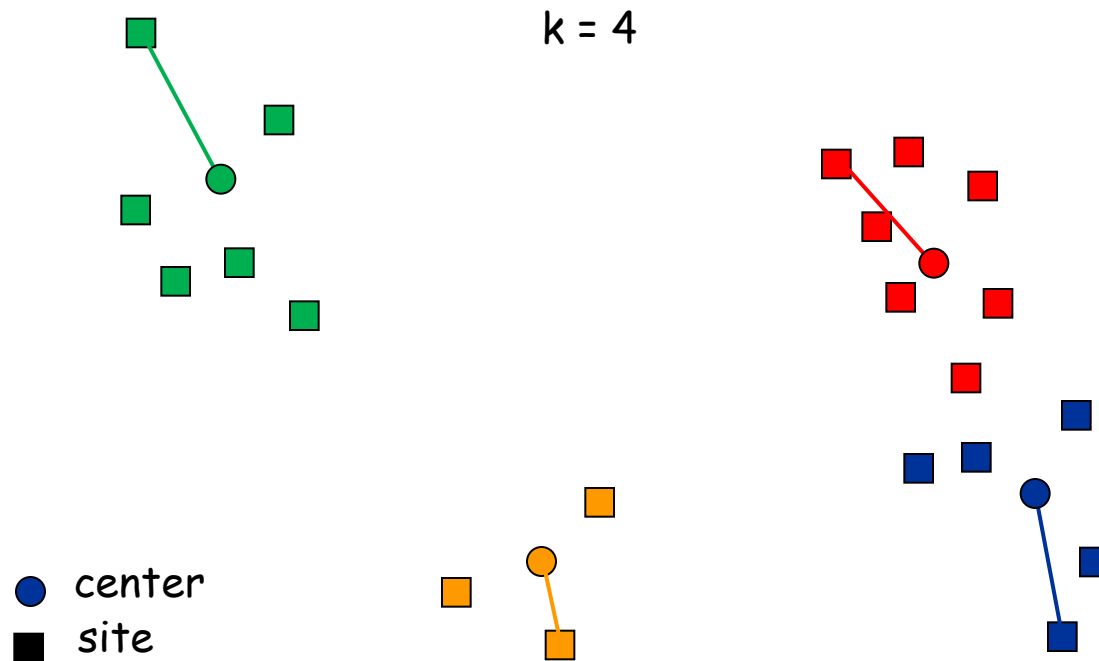
Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

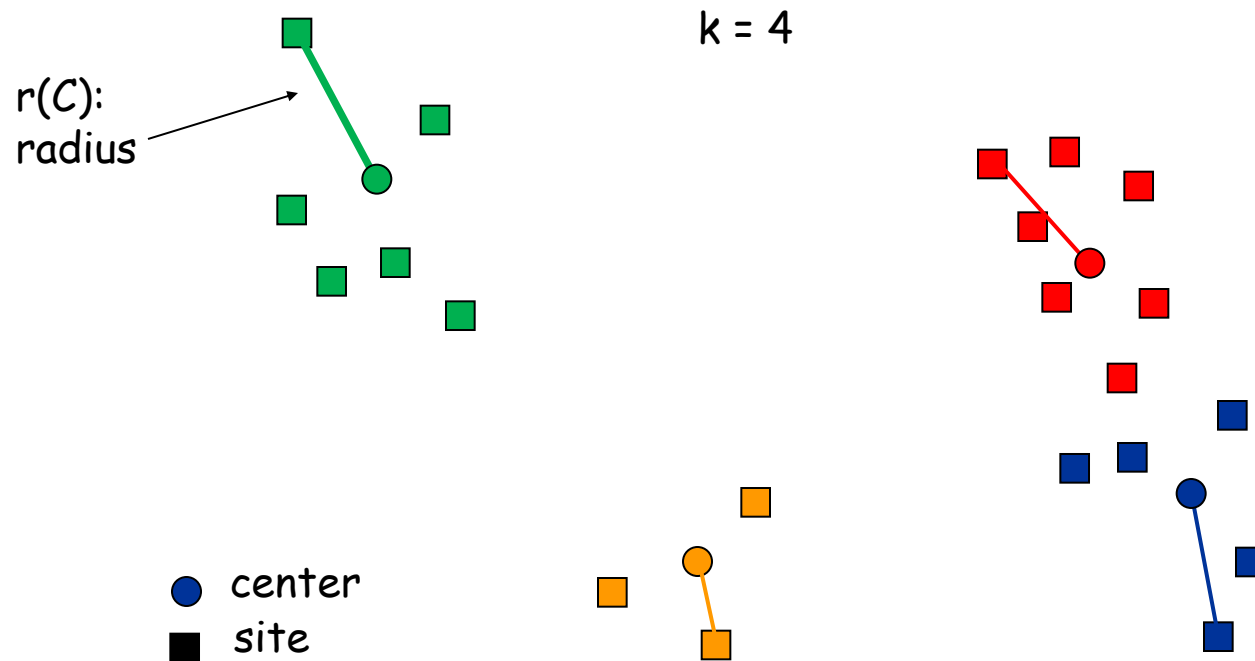
Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

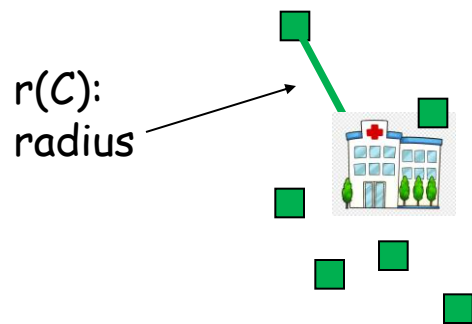
Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



k-Center Problem

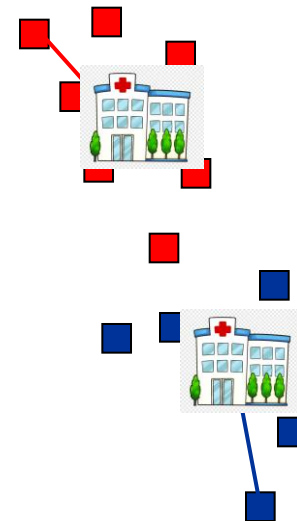
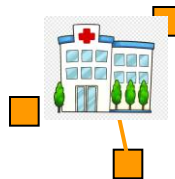
Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



$k = 4$

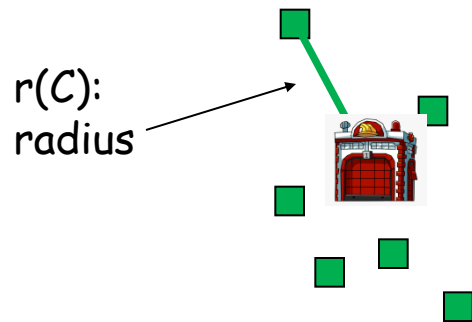
● center
■ site



k-Center Problem

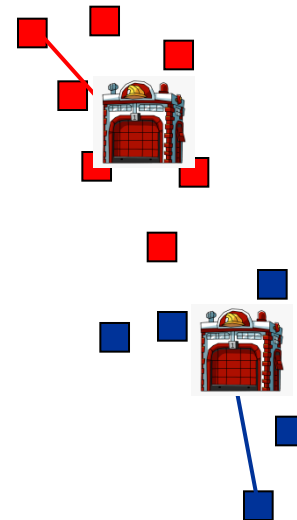
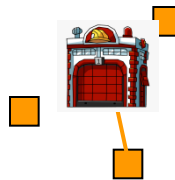
Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



$k = 4$

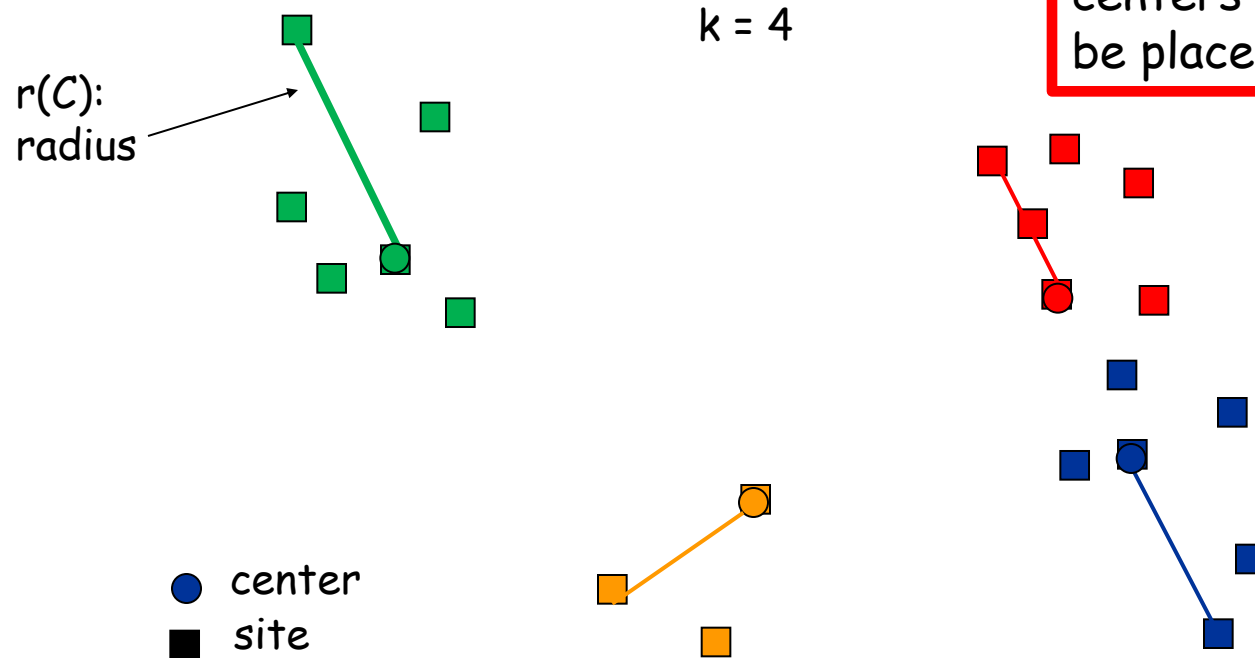
● center
■ site



k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.



k-Center Problem

Input. Set of n sites $s_1, \dots, s_n$ and integer $k > 0$.

Center selection problem. Select k centers C so that maximum distance from a site to nearest center is minimized.

Notation.

- $\text{dist}(x, y)$ = distance between x and y .
- $\text{dist}(s_i, C) = \min_{c \in C} \text{dist}(s_i, c)$ = distance from s_i to closest center.
- $r(C) = \max_i \text{dist}(s_i, C)$ = smallest covering radius.

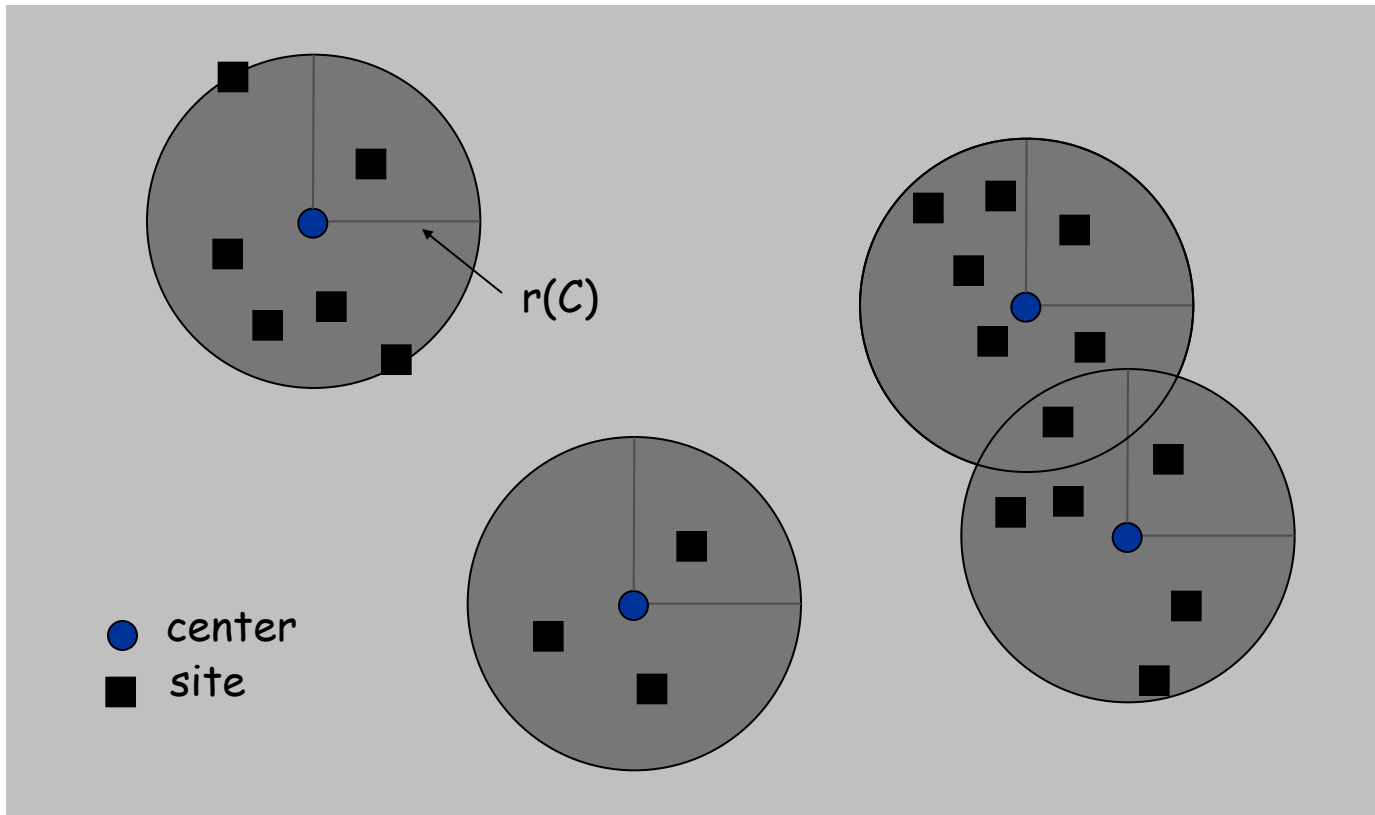
Goal. Find set of centers C that minimizes $r(C)$, subject to $|C| = k$.

Distance function properties.

- $\text{dist}(x, x) = 0$ (identity)
- $\text{dist}(x, y) = \text{dist}(y, x)$ (symmetry)
- $\text{dist}(x, y) \leq \text{dist}(x, z) + \text{dist}(z, y)$ (triangle inequality)

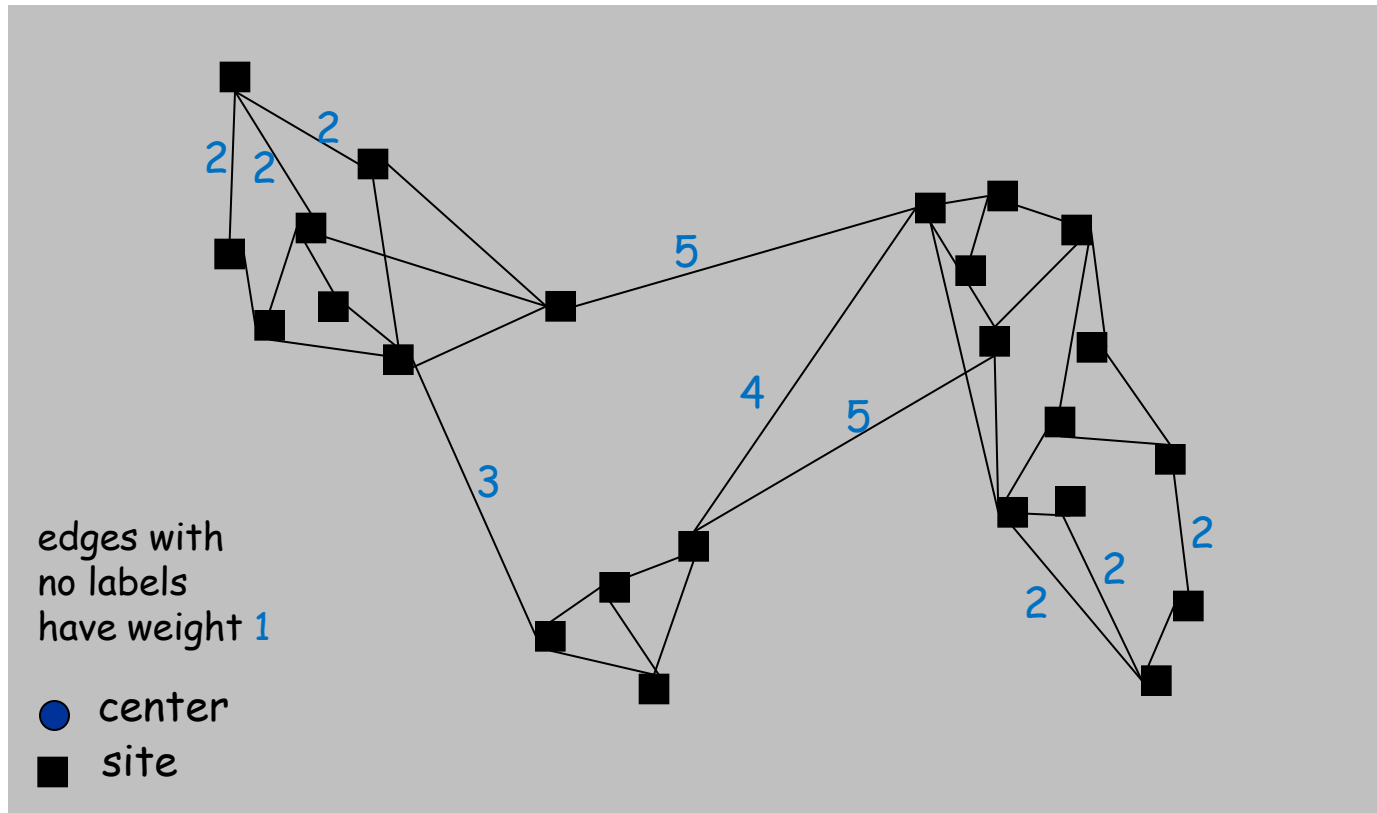
k-Center Problem

Ex: each site is a point in the plane, a center can be any point in the plane, $\text{dist}(x, y) = \text{Euclidean distance}$.



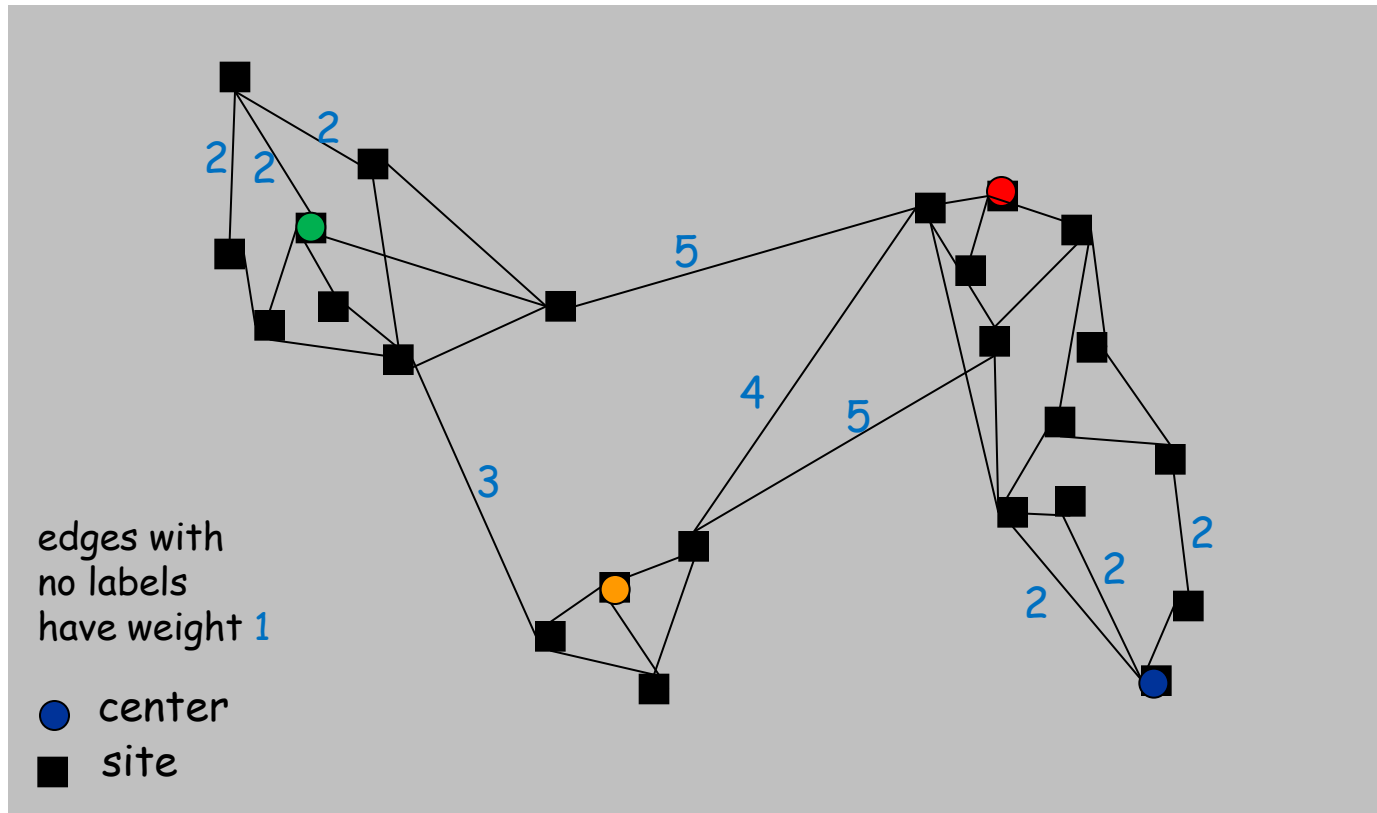
k-Center Problem

Ex: each site is a vertex in undirected weighted graph, a center can be any vertex, $\text{dist}(x, y) = (\text{weighted})$ distance in G between x and y .



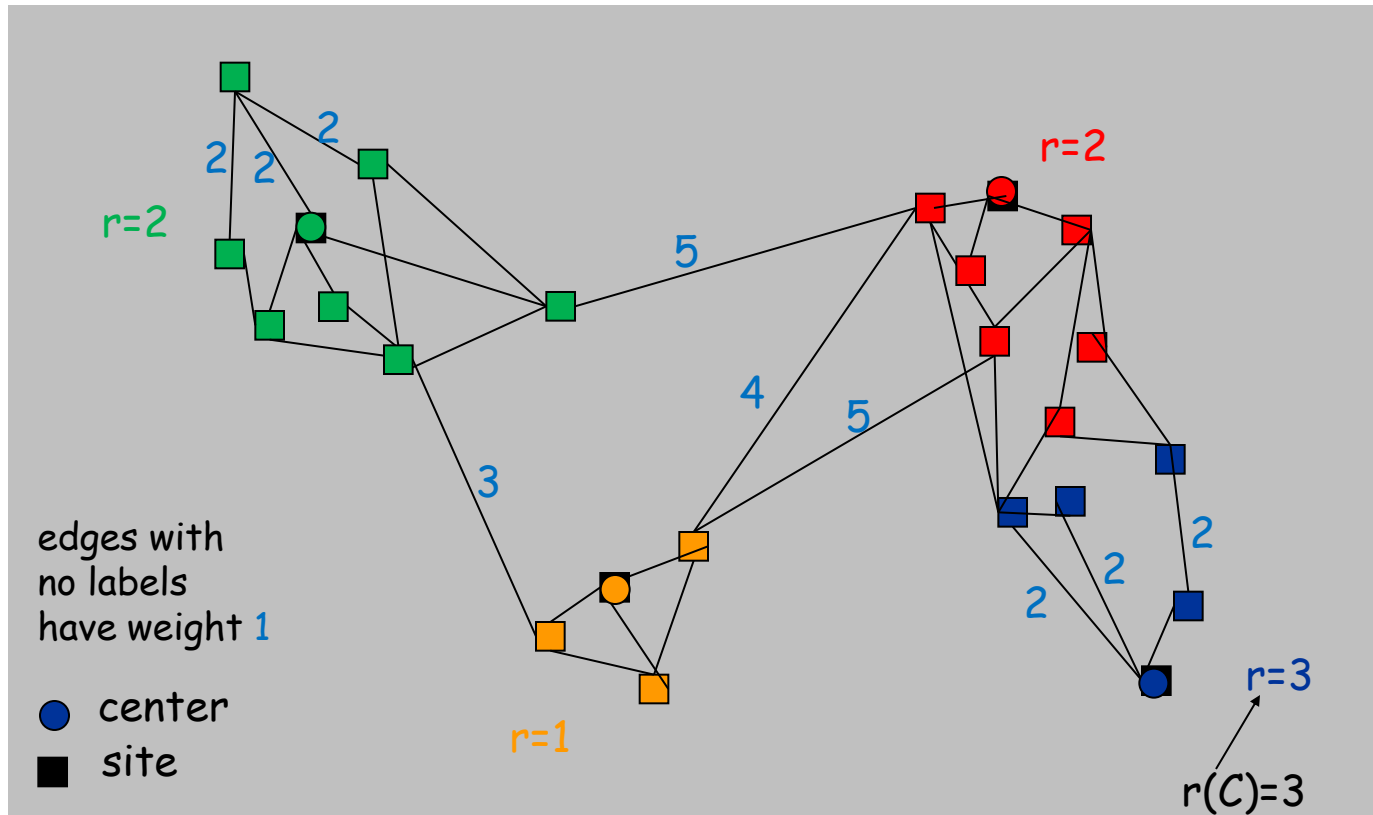
k-Center Problem

Ex: each site is a vertex in undirected weighted graph, a center can be any vertex, $\text{dist}(x, y) = (\text{weighted})$ distance in G between x and y .



k-Center Problem

Ex: each site is a vertex in undirected weighted graph, a center can be any vertex, $\text{dist}(x, y) = (\text{weighted})$ distance in G between x and y .



Center Selection: Greedy Algorithm

Greedy algorithm. Repeatedly choose the next center to be the site **farthest** from any existing center.

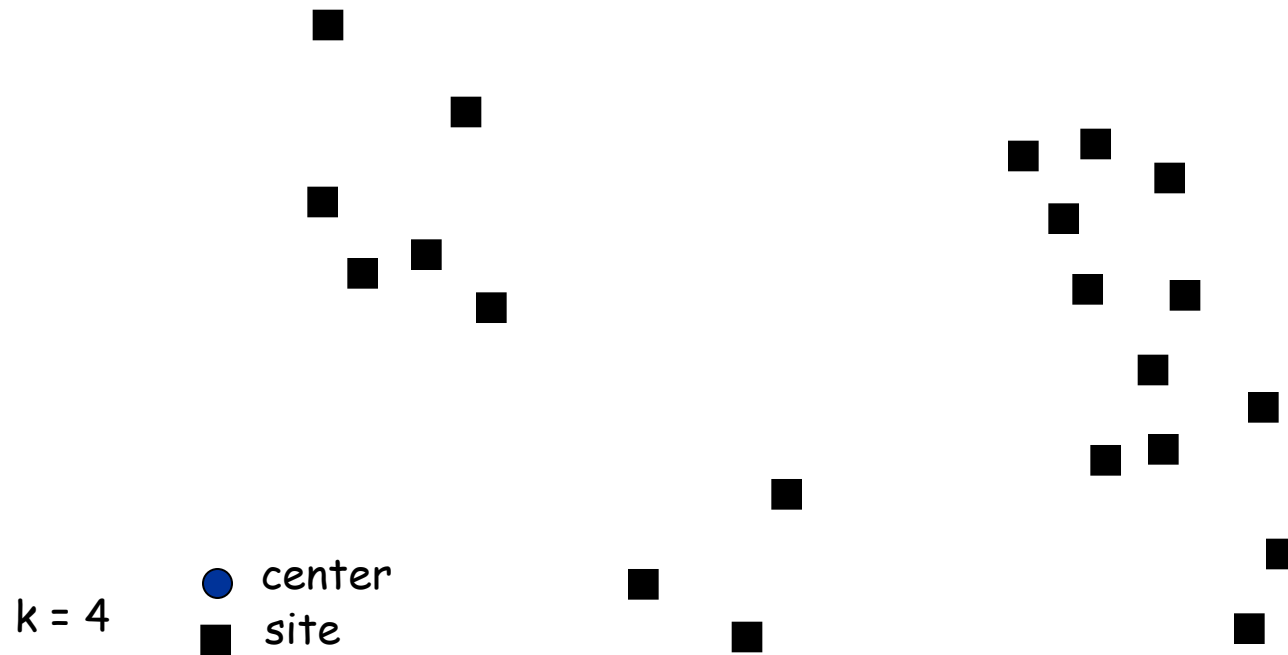
```
Greedy-Center-Selection(k, n, s1, s2, ..., sn) {  
  
    C =  $\phi$   
    repeat k times {  
        Select a site si with maximum dist(si, C)  
        Add si to C  
    }  
    return C  
}
```

↑
site farthest from any center

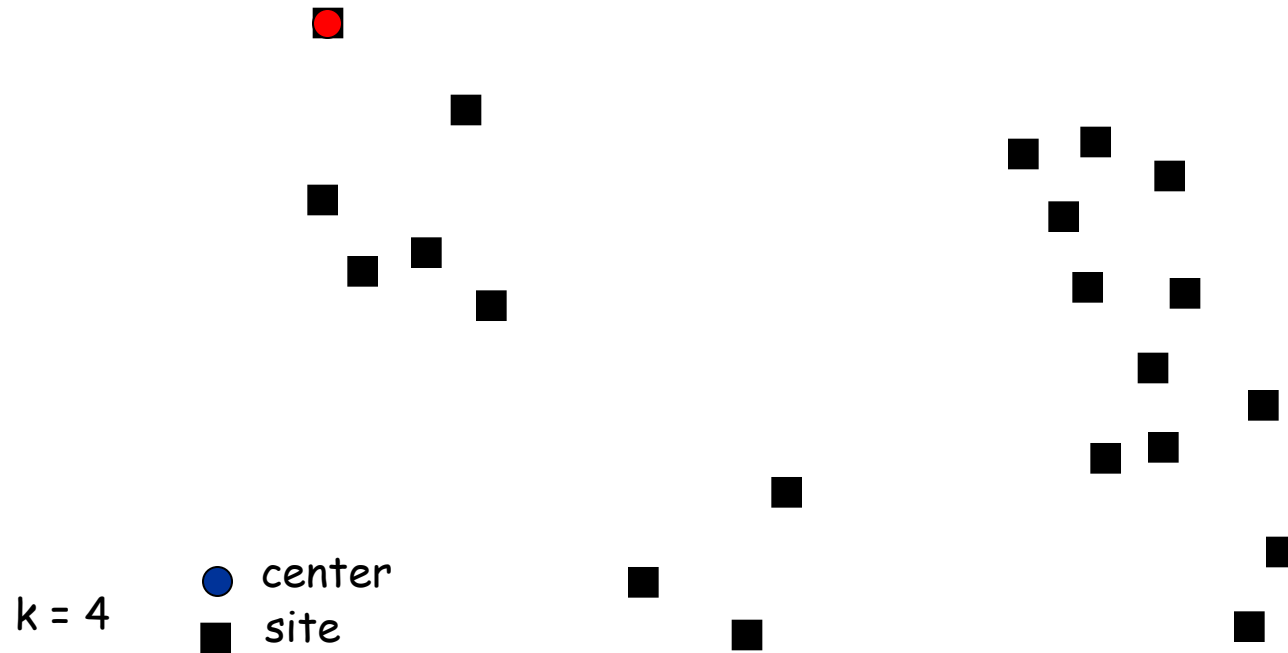
Observation. Upon termination all centers in C are pairwise at least $r(C)$ apart.

Pf. By construction of algorithm.

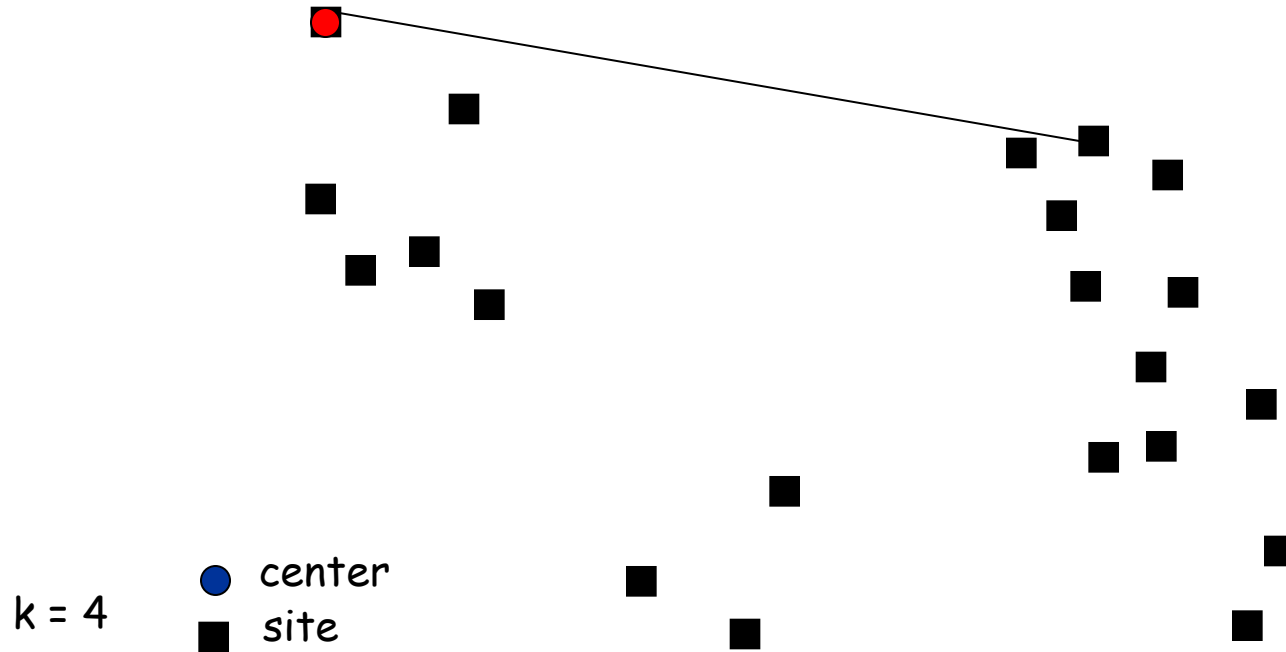
example of execution



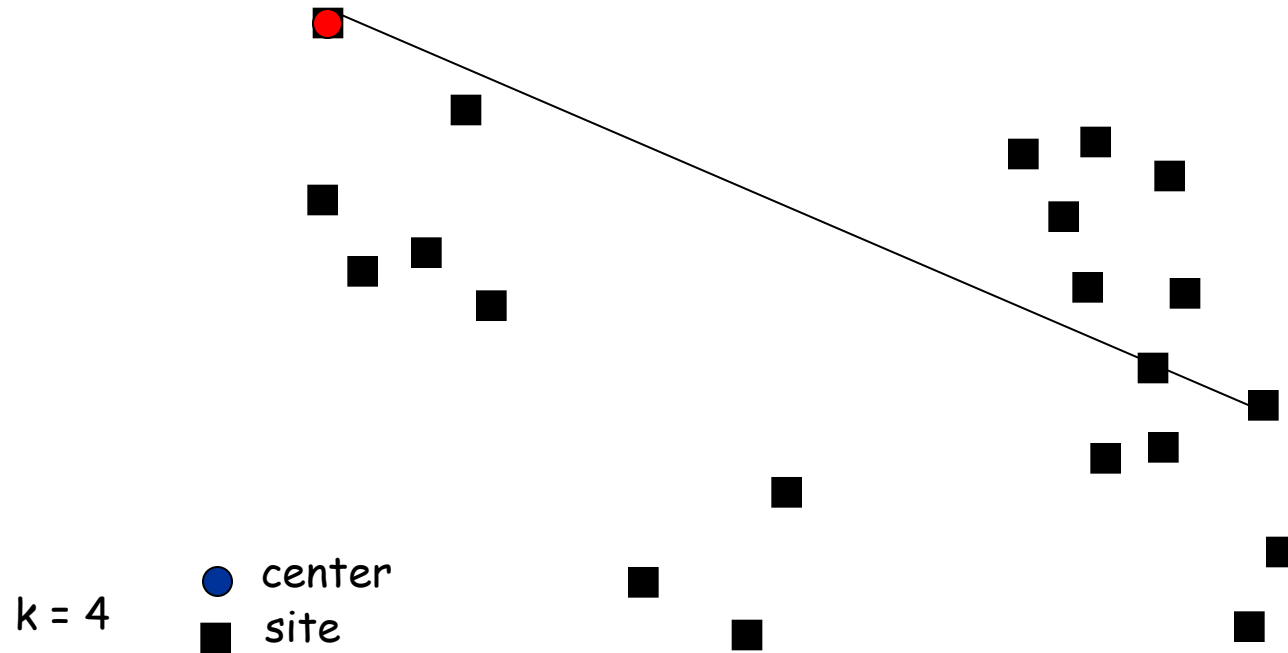
example of execution



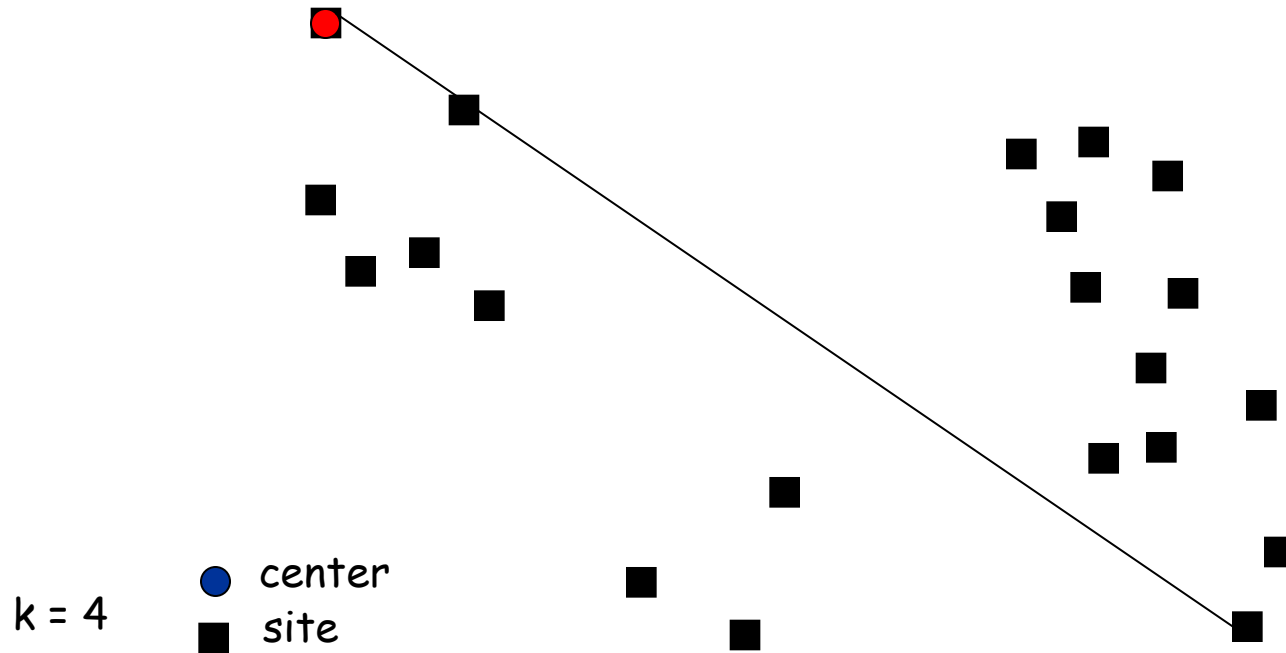
example of execution



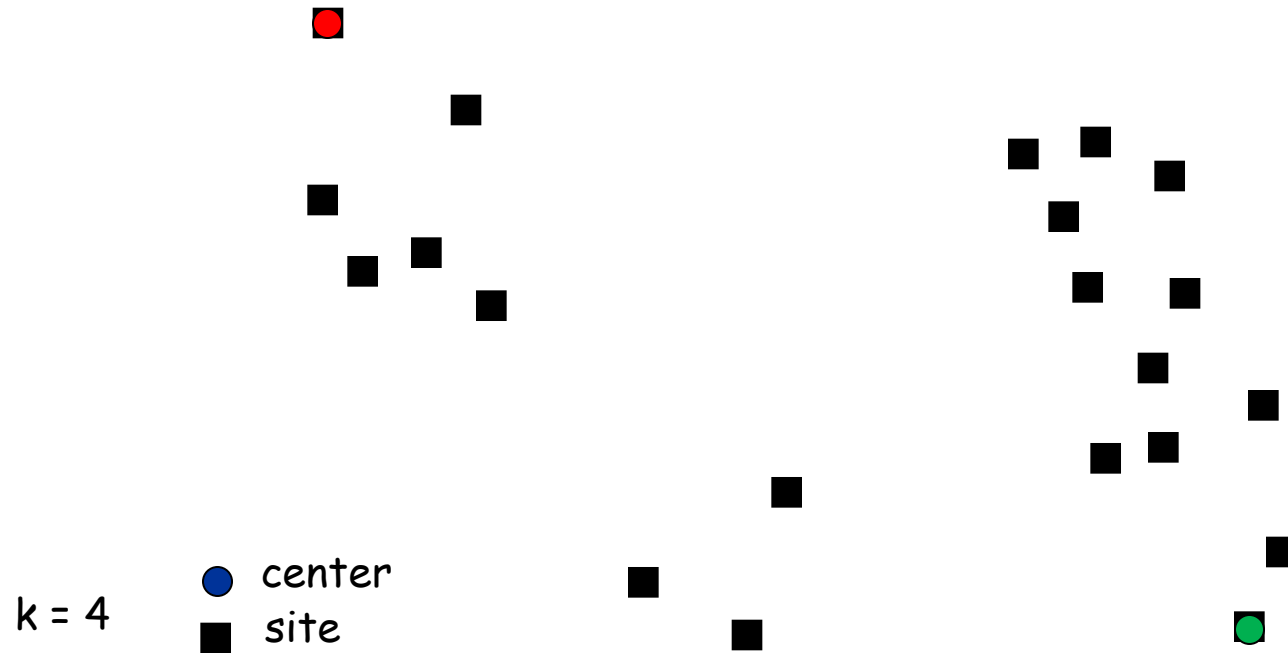
example of execution



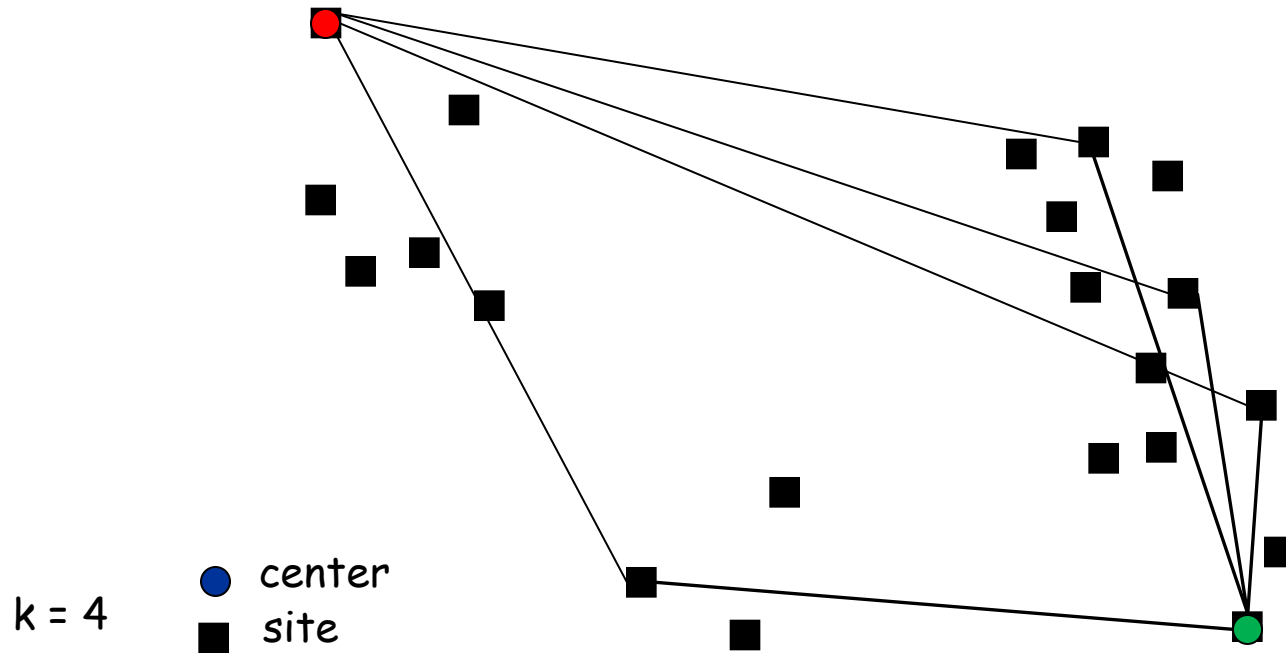
example of execution



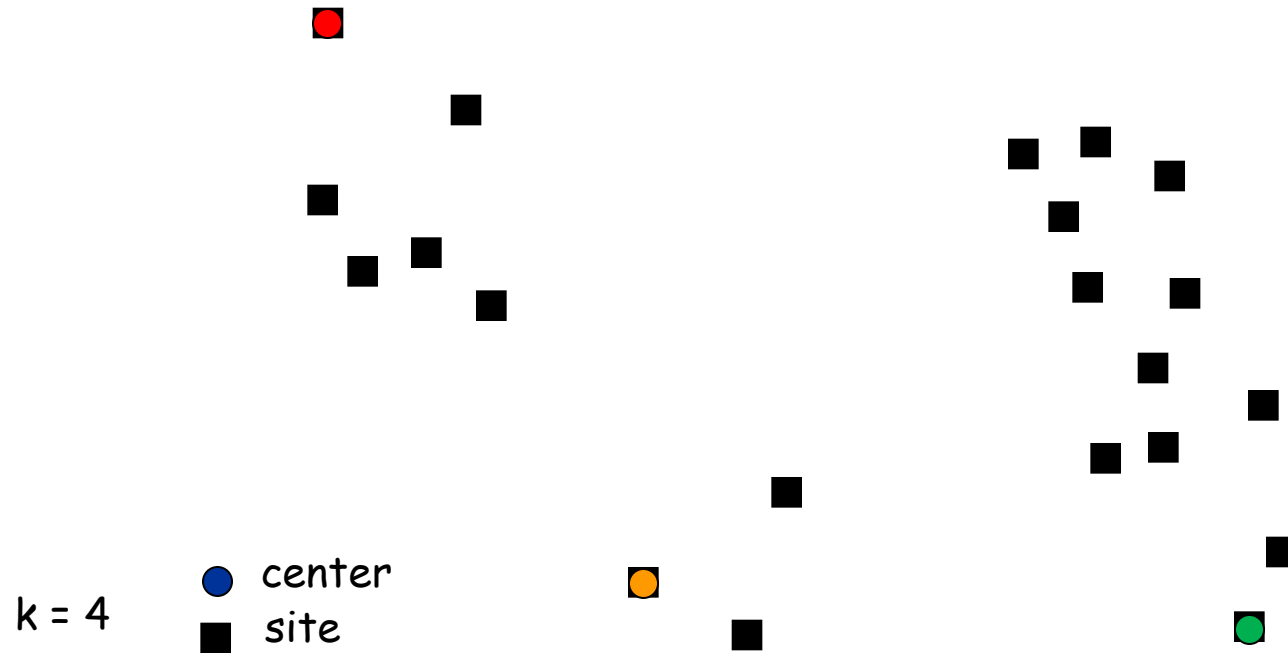
example of execution



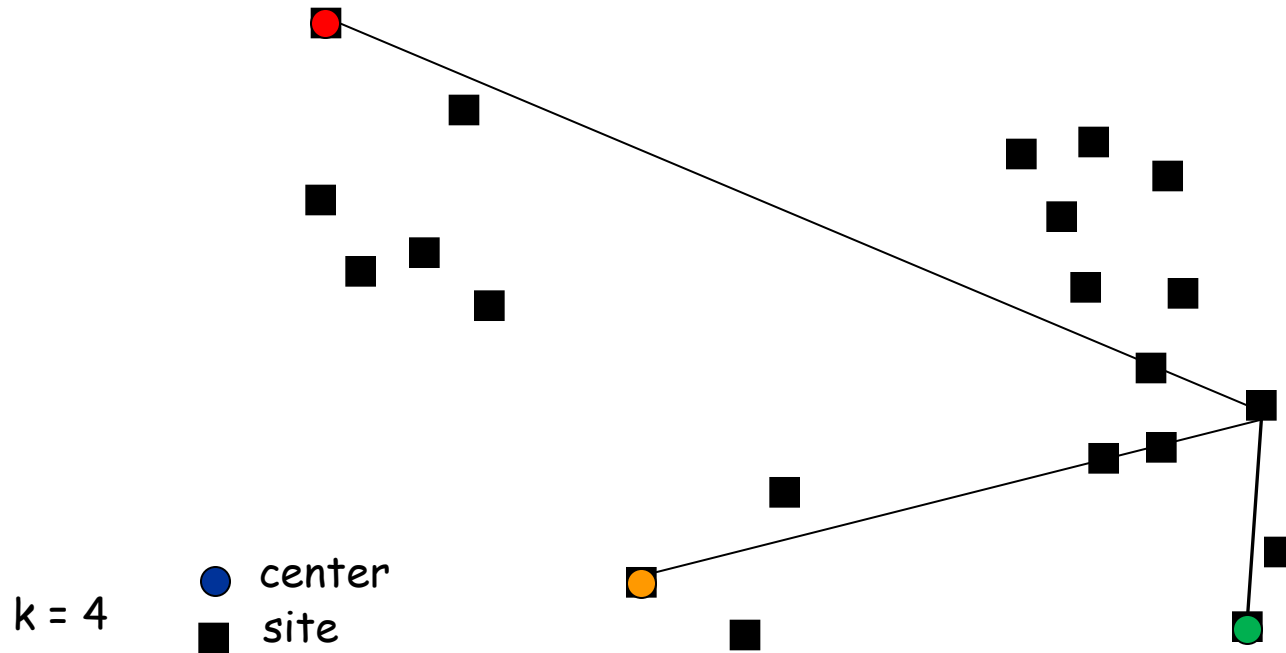
example of execution



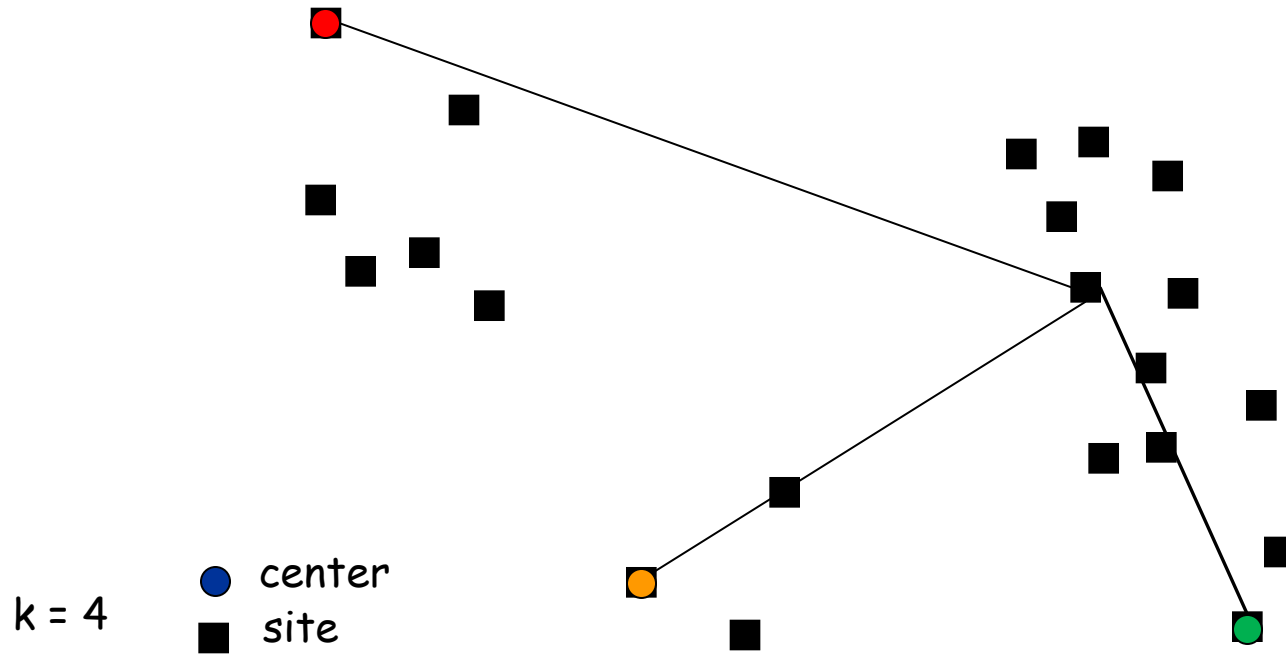
example of execution



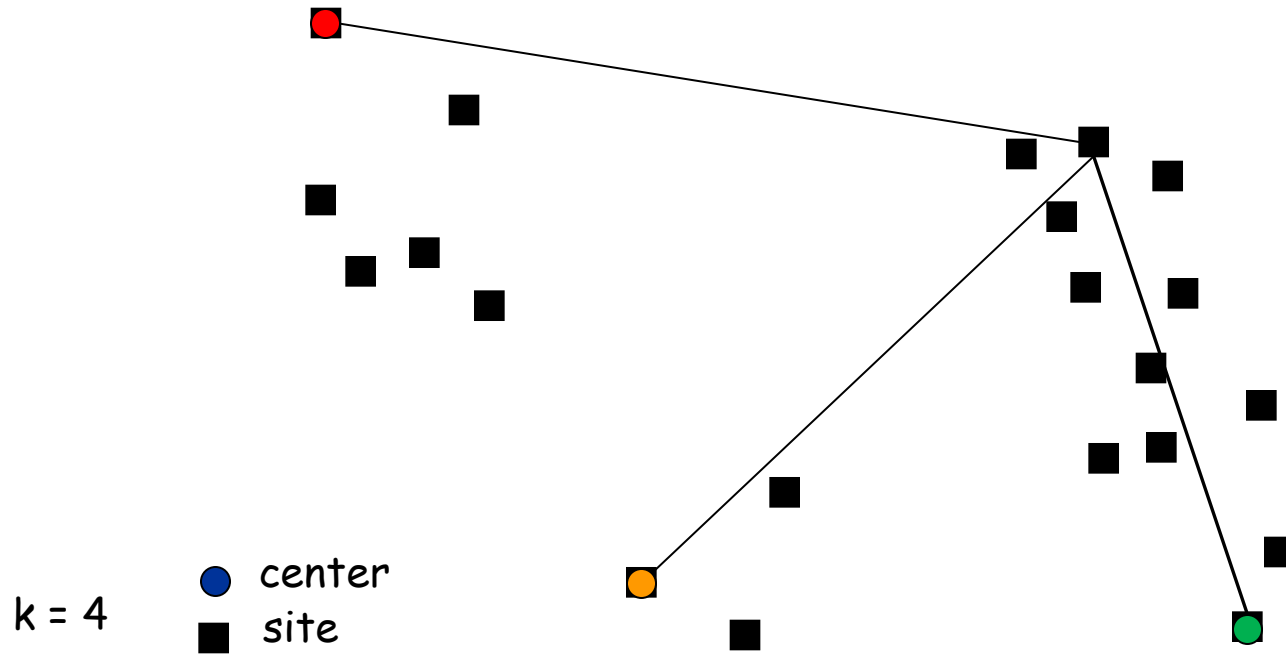
example of execution



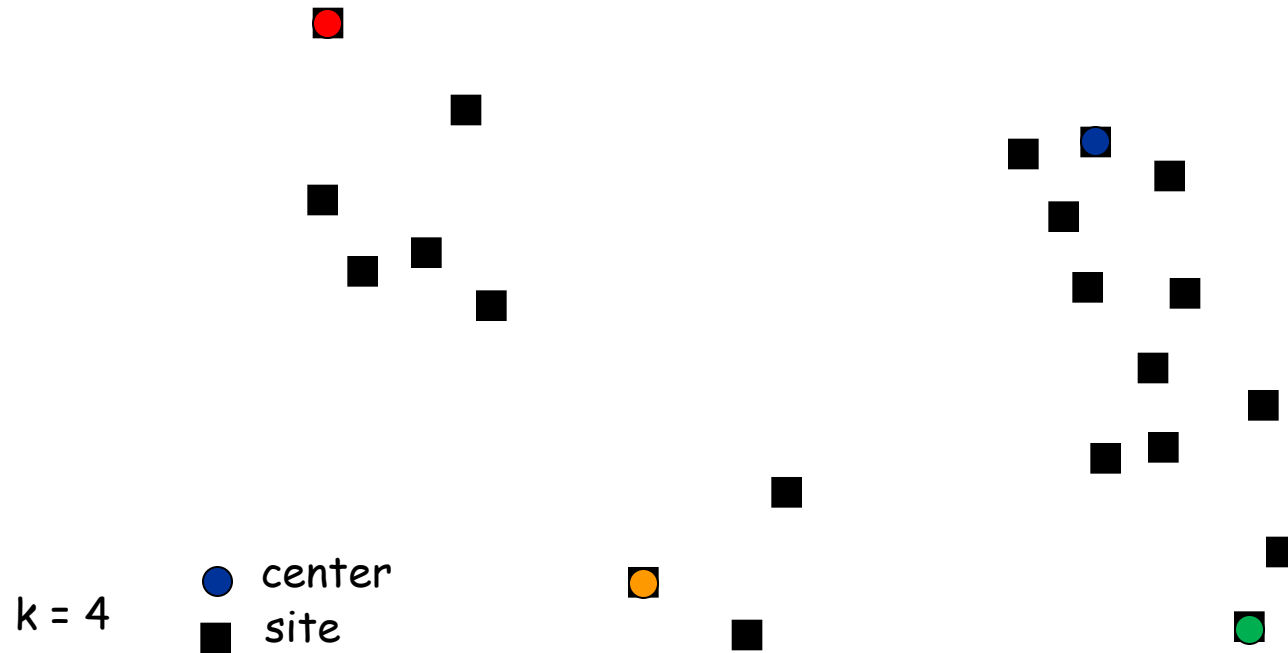
example of execution



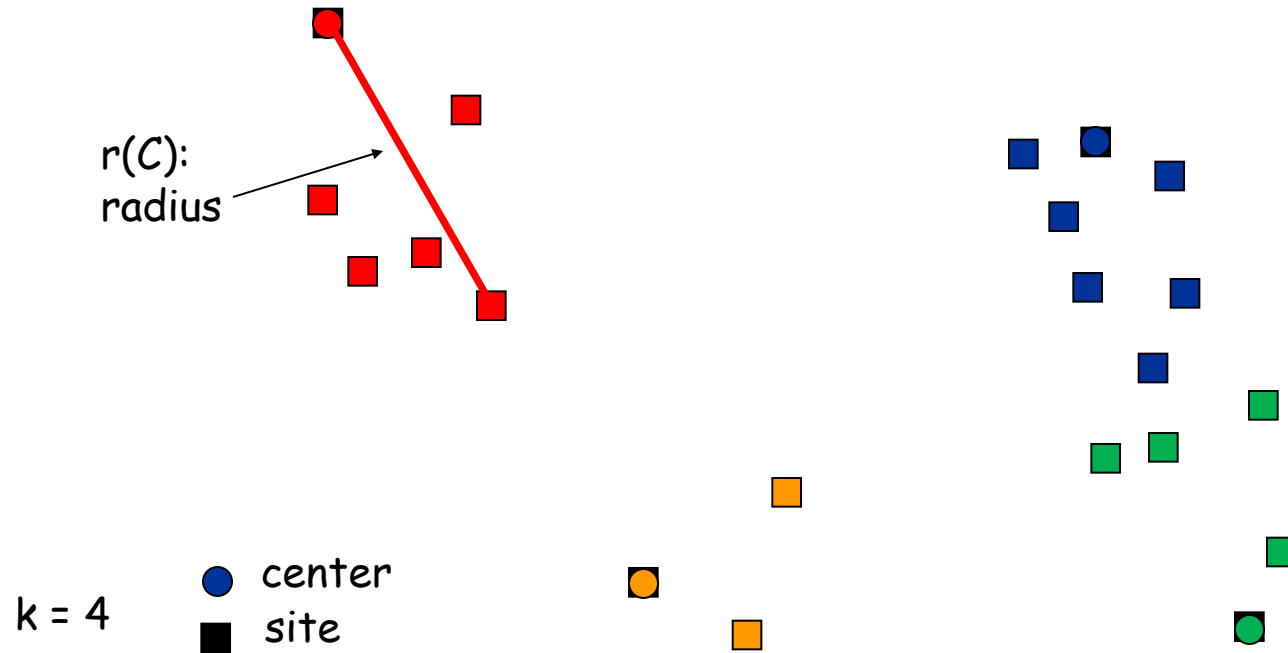
example of execution



example of execution



example of execution



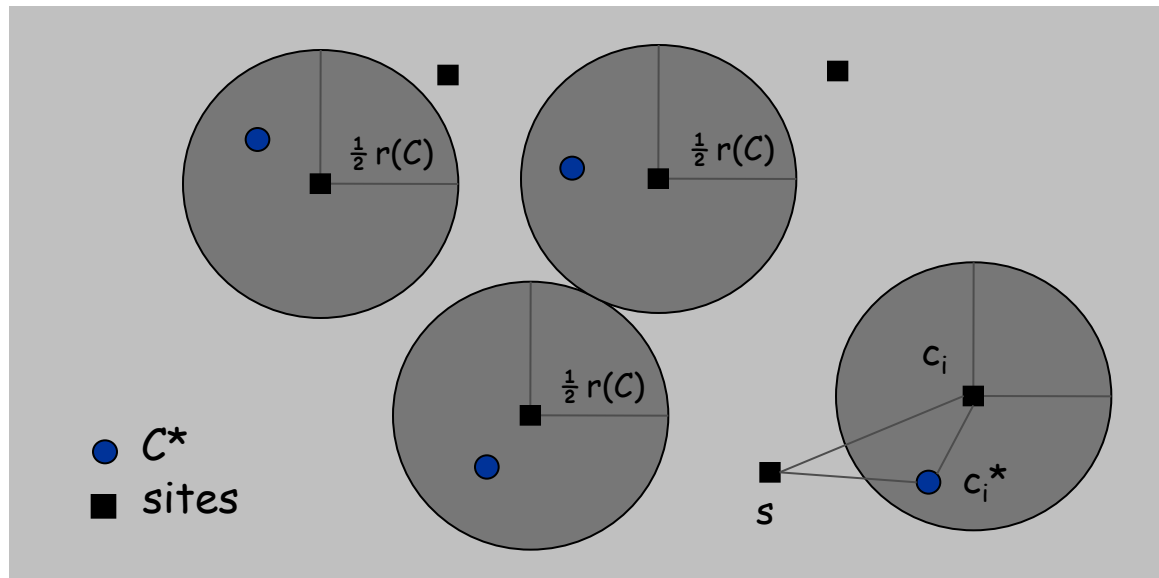
Center Selection: Analysis of Greedy Algorithm

balls are disjoint since all centers in C are pairwise at distance at least $r(C)$

Theorem. Let C^* be an optimal set of centers. Then $r(C) \leq 2r(C^*)$.

Pf. (by contradiction) Assume $r(C^*) < \frac{1}{2} r(C)$.

- For each site c_i in C , consider ball of radius $\frac{1}{2} r(C)$ around it.
- Exactly one c_i^* in each ball;
 - each ball with center $c_i \in C$ must contain a center in C^* (otherwise $\text{dist}(c_i, C^*) \geq \frac{1}{2} r(C) > r(C^*)$);
 - balls are disjoint and $|C| = |C^*|$.



Center Selection: Analysis of Greedy Algorithm

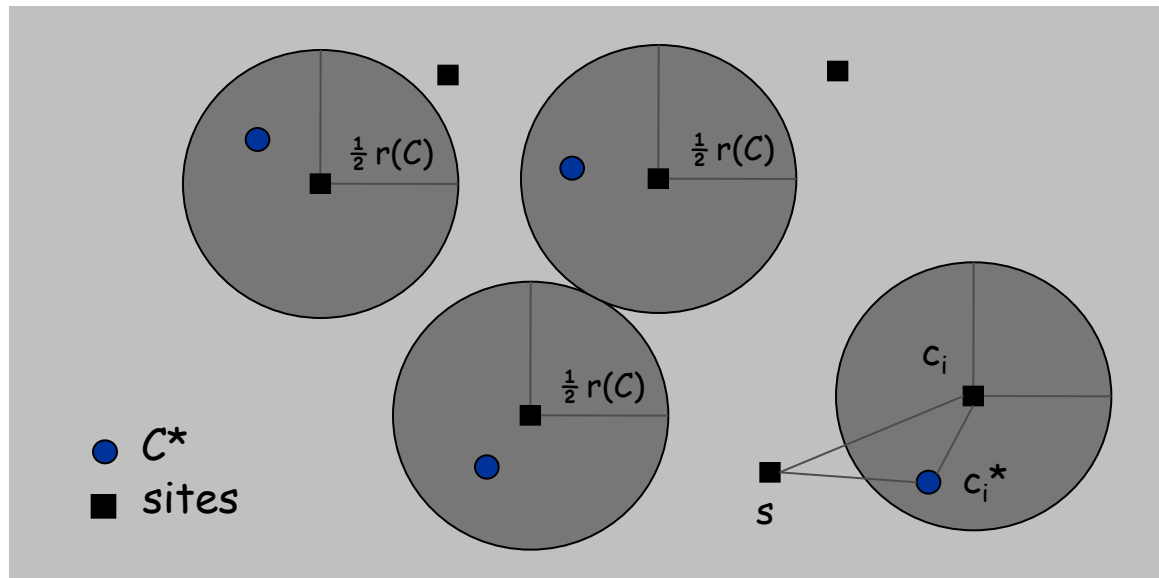
Theorem. Let C^* be an optimal set of centers. Then $r(C) \leq 2r(C^*)$.

Pf. (by contradiction) Assume $r(C^*) < \frac{1}{2} r(C)$.

- For each site c_i in C , consider ball of radius $\frac{1}{2} r(C)$ around it.
- Exactly one c_i^* in each ball; let c_i be the site paired with c_i^* .
- Consider any site s and its closest center c_i^* in C^* .
- $\text{dist}(s, C) \leq \text{dist}(s, c_i) \leq \text{dist}(s, c_i^*) + \text{dist}(c_i^*, c_i) \leq 2r(C^*)$.
- Thus $r(C) \leq 2r(C^*)$. ■

△-inequality

$\leq r(C^*)$ since c_i^* is closest center



Center Selection

Theorem. Let C^* be an optimal set of centers. Then $r(C) \leq 2r(C^*)$.

Theorem. Greedy algorithm is a 2-approximation for center selection problem.

Remark. Greedy algorithm always places centers at sites, but is still within a factor of 2 of best solution that is allowed to place centers anywhere.

↖
e.g., points in the plane

Question. Is there hope of a better approximation?

...very unlikely:

Theorem. Unless $P = NP$, there no ρ -approximation for center-selection problem for any $\rho < 2$.